Hidden Markov Model For Target Tracking With UWB Radar Systems

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***Abstract*—In this paper we demonstrate the application of Hidden Markov Models (HMM) for localization and tracking in ultra wide band (UWB) radar networks. To improve localization, a Voronoi region based approach is utilized to form a HMM for detection and tracking of mobile target. The observations used for the HMM localization are obtained from the power delay profile of the received signals. In UWB systems the use of entire power delay profiles instead of the total power only, allows to reach higher localization accuracy. This is due to the power delay profile being a measure of the power as well as the time of arrival. Simulation results suggest a performance gain of 7dB over the maximum likelihood estimation for localization in presence of path loss at intermediate values of signal to noise ratio (SNR).**

I. Introduction

In ultra wide band (UWB) systems,localization of moving targets is obtained through the measurement of propagation parameters related to the target location [5]. Parameter estimation is performed by exchanging radio signals with fixed interrogators placed at known locations. The typical propagation parameters that are used for this purpose are times of arrival (TOA), time difference of arrival (TDOA), angles of arrival (AOA) and received signal strength(RSS) [1]. The relationship between these parameters and target position are obtained either by analytical methods or through empirical measurements (eg. RSS profile maps). Estimation of target-interrogator distances and directions is achieved through these models. In a cognitive radar network (CRN) several radars co-operate together to provide the overall sensing system with the capabilities of (i) being aware of its outside world by using previous measurements as well as learning through interactions with the environment, and (ii) intelligently and adaptively customizing the operation of its transceivers in response to the channel variations in real time [13]. Two major cognitive mechanisms for the network have been suggested (i) distributed cognition, where the individual radars as well as the central control unit are all cognitive; and (ii) centralized cognition, where cognition including data fusion is confined to the central control unit, which in effect, acts as a brain of the entire network [14]. In this paper, we explore the potential of using the CRN for remote tracking of a targets movement within indoor environments - a key application suitable for cognition. Based on the readings collected from multiple interrogators , the location of the target can be estimated. The localization is based upon the triangulation approach. The choice of measurement type (e.g. TOA, TDOA, AOA, RSS) and the localization approach (e.g. centralized or distributed) depends upon the characteristics of application like indoor or outdoor. Errors in localization are caused due to parameter estimation errors faulty modeling, multipath effects and non line of sight conditions. In contrast to the band limited wireless systems UWB signals facilitate high resolution ranging possible [2,3,4]. In this paper we consider a UWB network with more than 3 interrogators placed in known positions and covering the area where the target has to be localized. Accurate ranging could be obtained, in theory by estimating TOA or TDOA from signals at the output of the matched filter, relying on high resolution obtained from the transmitted UWB pulse. However, dense multipath and large delay spreading , often found in indoor environments, affects the inherent resolution of UWB ranging systems. In this paper we use RSS-delay profile measurements instead of the conventional localization approach of estimation based upon RSS only. The target location is estimated by collecting all the signals collected upto the current time instant [5]. The estimation method is an adaptation of detection and tracking [6]. This algorithm can work in real time by maximizing the a posteriori probability of the hidden state given all the signals collected upto the current time instant. The algorithm can track by exploiting the continuity information in the trajectory. The power delay profiles for the signals received over each interrogator link are used to track the most likely sequence. It is worth noticing that, unlike other bayesian estimators such as Kalman Filter(KF) or the extended kalman filter (EKF) [7],[8], this HMM based algorithm does not rely upon the linearization and gaussian assumptions, still preserving about the same computational complexity of the above mentioned algorithms. The key contributions of this paper are:

1) Developing a Hidden markov model based on topological properties of the region in the UWB radar framework.
2) Comparison between the maximum-likelihood estimates of the target position and estimates of the target location obtained through the proposed HMM model.

The proposed research aims at improving the localization accuracy through use of HMM. In section II we present an introduction to HMM model and describe the approach used for detection of the target and RSS-TOA based analysis for localization. In section III we analyze how a comparative study

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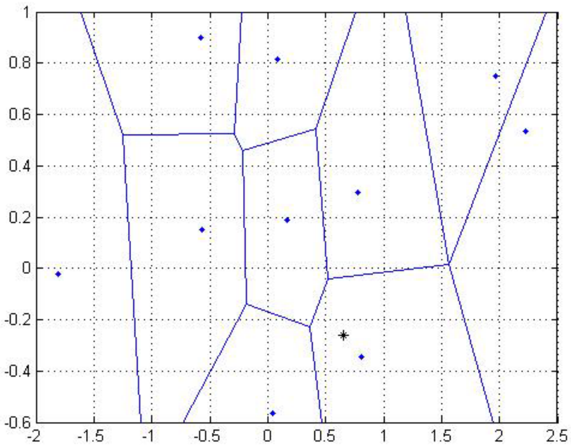

Fig. 1. Voronoi Regions

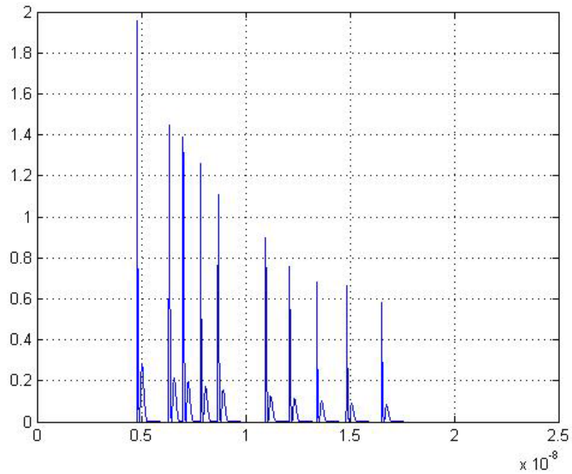

Fig. 2. Received first arrived powers at each of the interrogators

between the maximum likelihood estimation of the position and HMM can be made. In section IV we provide simulation results for validating the accuracy of the model. In section V we provide discussion on results. At the end of this paper we provide a set of conclusions in section VI.

II. System Model

A Voronoi region based approach is adopted to develop a hidden Markov model for detection and tracking of the target. We classify the Markov states as the independent Voronoi regions in the geography, the purpose behind utilizing the Voronoi regions was to enable classification of the states in the proposed HMM model and also to facilitate the localization of the target by a set of three interrogators in the vicinity. . The target is assumed to be equi-probable anywhere in the geography or in other words any state. It is also assumed that the target makes random walks in the 2-D geometry. The target is driven by a random process which is assumed to be Gaussian. The interrogators in the region are assumed to be randomly distributed. The region is then divided into distinct Voronoi regions with each interrogator acting as the Voronoi centre. As indicated in Fig.1, the initial target location is assumed to be equiprobable in any of the states, as described earlier the target makes transition from one state to other based upon the random perturbation.

$$\tilde{x}_{\mathrm{t}+1} = \tilde{x}_t + v \tag{1}$$

where v is a random perturbation and $\tilde{x}$ denotes the position vector. The transition probability and the emission probability parameters of the Hidden Markov process are optimized using the Baum-Welsh algorithm to fit the sequence of observations as indicated in [9]. The target is driven by a random process over the region under consideration with randomly distributed interrogators. In this paper a comparative study is made between the maximum likelihood estimation and the estimation provided by the hidden markov model. The proposed algorithm assumes that the localization problem under consideration adopts a centralized approach, i.e. the RSS-TOA profiles from the set of interrogators is relayed to a central data fusion centre which is then responsible for developing a Markov model on the basis of the observations up to the current instant. The transmitted UWB radar pulses are assumed to be the second derivative of Gaussian pulses, the power delay profile is as shown in Fig.2 The first arrival delay equals the propagation time $\tau = \frac{d}{(c\delta t)}$ over the target interrogator distance d, where c represents the velocity of light and δt represents the excess delay. To account for the dependence of the RSS on the propagation distance d , the first arrival power is given by σ_z^2

$$\sigma_z^2 = \sigma_{\mathrm{ref}}^2 (\frac{d}{d_{\mathrm{ref}}})^{-\alpha} = \sigma_{\mathrm{ref}}^2 (\frac{\tau}{\tau_{\mathrm{ref}}})^{-\alpha}$$

where σ_{ref}^2 represents the reference power at a reference distance and α represents the the path loss exponent (typical values of $(\alpha = 2 - 8)$. The RSS-based localization accuracy suffers from shadowing fluctuations. The effects of random shadowing on indoor localization have been investigated by authors in a wireless sensor network scenario [10]. The localization approach considered is a version of HMM method based on RSS measurements only. In this paper both RSS and power delay profile measurements are considered, but shadowing is not included in the channel model. The extension to shadowing channels is possible by modeling $\sigma_z^2(\tau)$ as a log-normal random variable and modifying the distribution of the measured signal y which represents the received pulses in the power delay profile. Propagation models including shadowing effects for power delay profile based localization can be found in [11] for indoor and [12] for outdoor environments. The basic purpose behind HMM algorithm is target localization at each time instant i; this can be achieved considering not only the current measurement y_i but all the measurements $y_{0:i}$ accumulated over the path of the target up to the current instant. The HMM state is defined as the Voronoi region the target belongs to as $q_i = x_i \in \chi$ where χ indicates the entire region under consideration. This state is hidden in the interrogator link observation vector y_i given as

$$y_i = f(x_i) + \eta_i \tag{2}$$

where $f(.,.)$ denotes the vector of non linear functions indicating the relation between the location state q_i and the signal , while $\eta_i = [\eta_{i,1}^T, ..., \eta_{i,N}^T]^T$ is the noise vector. The target position is modeled as first-order homogeneous Markov chain (MC). In order to characterize the Markov chain $x_i \sim MC(\pi, A)$, we need to define the initial state probabilities $\pi = [p(x_0)]$ and the transition probabilities $A = p(x_i|x_{i-1})$. Since the Markov state is hidden in the observations y_i, we also have to define the emission/observation sequence pdf $B = p(y_i|x_i)$; these probabilities compose the overall HMM parameter set named $\lambda = (A, B, \pi)$, where A indicates the transition probability matrix, B indicates the emission or observation probability and π indicates the initial state probability. Within the 2-D space, the target trajectory which is a direct result of the homogeneous Markov chain $q_i \sim MC(\pi(q), A(q))$ according to the random process as described earlier $q_i = q_{i-1} + v_i$ where v_i is the 2-D driving process with known distribution $f_i(x) = p(v_i = x)$. The transition probabilities $A_q = [a_{\mathrm{m,n}}^q]$ are calculated as :

$$a_{\mathrm{m,n}}^q = p(q_i = n|q_{i-1} = m) \propto f_v(n - m) \tag{3}$$

for $m, n \in \chi$, the form of distribution $f_v(n)$ is closely related to the features of the target trajectory. The initial -state distribution is defined by designating the probabilities for the position π .i.e. by assigning the probabilities for the position $\pi(q)$ and the null state $p(x_0) = 0$ at time instant $i = 0$. A logical assignment in the case of missing a-priori information might be to assign equal probabilities to all possible states $p(x_0) = 1/\chi$. Thus the emission or observation pdf can now be defined as a pdf set B conditioned to $x_i = 0$

$$p(y_i|x_i = 0) = \prod_{\mathrm{l=1}}^{N} \Lambda(y_{\mathrm{i,l}}, \tau)$$

We will consider the estimation of the state sequence $x_{0:\mathrm{N}-1} = [x0, ..., xN - 1]$ from the observations $y_{0:\mathrm{N}-1}$ under assumption of known HMM set λ in Section III.

III. MAXIMUM LIKELIHOOD ESTIMATION AND HMM MODEL

In this section we consider the estimation of x_i from the signals measured only at the generic time instant (i.e.,local estimation only) using the maximum likelihood estimation(MLE) approach.

A. MLE approach

The local MLE from the interrogator measurement y_i is obtained by maximizing the likelihood function approach. The local MLE from the interrogator - target link measurement y_i is obtained by maximizing the likelihood function The local MLE from the interrogator measurement y_i is obtained by maximizing the likelihood function approach. The local MLE from the interrogator - target link measurement y_i is obtained by maximizing the likelihood function

$$x_i^{\mathrm{mle}} = \arg\max p(y_i|x_i) \tag{4}$$

the observations $(y_{(i,l)})_{l=1}^N$ conditioned to the position x_i, hence the likelihood function simplifies to the product of marginal probabilities for each link.

$$p(y_i|x_i = k) = \prod_{\mathrm{l=1}}^{N} p(y_{\mathrm{i,l}}|q_i = x) \tag{5}$$

where y_i is the observation vector or the RSS-TOA profile where $q_i \in \chi$ where χ belongs to the entire region under consideration, N indicates the total number of interrogators. According to Gaussian assumptions for the signal model, the conditioned probability given the LOS delay τ is given as mentioned in [5]:

$$\Lambda(y, \tau, \delta\tau) = |C(\tau, \delta\tau)|^{-0.5}(2\pi)^{-0.5k} exp[-0.5y^T C^{-1}(\tau, \delta\tau)y]$$

$\delta\tau$ represents the excess delay in the case of non-line of sight conditions. $C(\tau, \delta\tau)$ represents the covariance of delay and excess delay parameters. Thus for l interrogators the l^{th} likelihood function $p(y_{\mathrm{i,l}}|x_i)$ can now be easily re-written as a function of $\Lambda(y, \tau, \delta\tau)$. Thus it can be simplified as

$$p(z_{\mathrm{i,l}}|x_i) = \Lambda(z_{\mathrm{i,l}}, \frac{||x_i - x_l||}{c}) \tag{6}$$

B. HMM approach

By utilizing different estimation methods from the HMM theory given the model [9], the most optimum state sequence $x_{0:N-1}$ associated with the ordered measurement set $y_{0:N-1}$. Global criterion based methods are used to estimate all states $x_{0:N-1}$ from the entire set of observations $y_{0:N-1}$. Maximum aposteriori state by state estimation can be achieved by maximizing for each state x_i the aposteriori pdf $p(x_i|y_{0:N-1}, \lambda)$ evaluated through the forward backward algorithm. Alternatively, the Viterbi algorithm provides a method to select the optimum state sequence $x_{0:N-1}$ which maximizes $p(x_i|y_{0:N-1}, \lambda)$.

Due to large computational complexity and latency in state estimation both backward-forward algorithm and Viterbi algorithm are not suitable for real time localization. Therefore there is a necessity to consider localization as a forward only process that estimates x_i based on all the measurements $y_{0:N-1}$ collected up to the i^{th} time instant. The detection and tracking algorithm is a Bayesian approach developed by authors for UWB radar system [10] and used in other different applications [11]. Here the method is adopted to estimate the position state $x_i = q_i$ by maximizing over the entire region χ, the posteriori pdf $(x_i = p(x_i|y_{0:i}), \lambda)$ given the measurements $y_{0:i}$ up to the current time instant, a similar approach has been adopted in [5] in order to estimate the location of mobile terminals in indoor localization scenarios.

$$\tilde{x}_i = \arg\max_{x_i \in \chi} \gamma_i(x_i) \tag{7}$$

The posterior pdf at time i is evaluated using the Bayes theorem.

$$\gamma_i(x_i) = \mu_i p(z_i|x_i) p((x_i|z_{0:N-1}), \lambda) \quad (8)$$

where $\forall, i = 0, 1, ..., N-1$, the normalization term μ_i is defined such that $\sum_{x_i \in \chi} \gamma_i(x_i) = 1$. The conditioned probability $p(y_i|x_i)$ is obtained from the current measurement vector y_i, while updating probability $p(x_i|y_{0:N-1}, \lambda)$ is calculated from the aposteriori pdf at the previous instant

$$\gamma_{i-1}(x_{i-1}) = p((x_{i-1}|z_{0:N-1}), \lambda) \quad (9)$$

throughout the transition probabilities of the Markov chain.

$$p((x_{i-1}|y_{0:N-1}), \lambda) = \sum_{x_{i-1} \in \chi} p(x_i|x_{i-1}) \gamma_{i-1}(x_{i-1}) \quad (10)$$

On the basis of the (7),(8)we can obtain the forward recursions that allows us to compute $\gamma_i(x_i)$ for all instants from $i = 0$ up to $i = N-1$. In the first instant, the aposteriori pdf is initialized by using the a priori distribution $p(x_0)$. The transition matrix A_q is largely sparse due to the dynamic nature of the model assumed for target motion. In fact, this matrix depends on the 2-D filtering kernal $f_v(n)$ that has a spatial support which is limited[12]. In realistic scenarios, only a partial a-priori knowledge of the HMM parameter set $\lambda = [\lambda, A, B]$ is available. A training procedure has to be adopted for optimally adapting the model λ to some observed data $y_{0:N-1}$ of length N given the model λ, in practical systems. A method to analytically derive the maximum likelihood estimate

$$\lambda^* = \arg\max_{\lambda}(p(y_{0:N-1}|\lambda))$$

is known. We can select λ for local maximization of the likelihood function $p(y_{0:N-1}|\lambda)$ through an iterative procedure in the HMM literature known as Baum-Welsh algorithm [9]. It is an expectation maximization (EM) technique that, starting from an estimate λ^j at iteration j, evaluates the aposteriori probabilities of the state transition, given the observation sequence $y_{0:N-1}$. These aposteriori pdfs achieved by assuming $\lambda = \lambda^j$, are then used to re-estimate the HMM parameters by approximating the probabilities constrained in λ in terms of expected frequencies of state transitions. The new parameter set λ^{j+1} is such that

$$p(y_{0:N-1}|\lambda^{j+1}) \geq p(y_{0:N-1}|\lambda^j)$$

The process terminates at the parameter set λ_{EM} when the convergence is reached or some limiting criterion are met. Global convergence is not guaranteed since this is a local algorithm and the solution quality depends on the selected initial parameters.

IV. DETECTION AND TRACKING SIMULATION RESULTS

In this model for localizing the target every parameter of the HMM model is subject to the estimation process. As it will be apparent from the simulation results that the performance of the system are insensitive to large variations of delay τ_{rms} and SNR η. We thus estimate the transition probabilities that compose matrix A or, equivalently, the pdf $f_v(n)$. To efficiently adjust the HMM parameters, the statistics of the target position process during the localization phase must be the same of those observed during the training phase. As simulation results show the target can be efficiently tracked even if there are slow or fast variations, provided that a similar behaviour has been observed during the training phase in order to efficiently adjust the transition probabilities and allow accurate tracking of the target. We now make a comparison between the maximum likelihood estimation process and the proposed HMM model in order to track the trajectory of the target in the Voronoi regions. As indicated in Fig.3 the error in the estimation of the amplitude of the received radar pulses through maximum likelihood estimation and the HMM model is as shown which clearly indicates the capability of the HMM model to effectively track the amplitude of the first arrival power of the UWB pulse transmitted by the interrogators. The deviation of the estimated value from the actual value is less relative to the one predicted by the MLE model, at a given value of SNR.

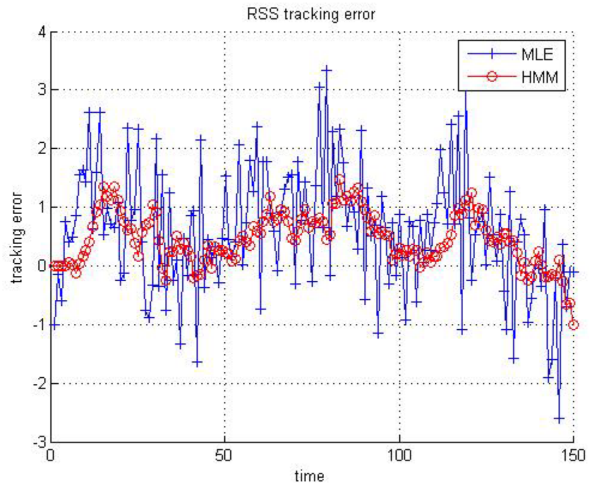

Fig. 3. Tracking error

V. DISCUSSION ON RESULTS

Measurements are generated according to the signal model as described earlier with a length of 400 samples. The UWB radar pulses are at a frequency of 4 GHz and the sampling frequency used is $f_s = 10$ GHz and the power delay profile is as indicated in Fig.2 with exponentially decaying power characterized by delay $\tau_{rms} = 10$ nS. Measurements are sampled at a frequency of 10 GHz, have length $L = 150$, with the first arrival delay $\tau_{i,l}$ being obtained from the interrogator target distance as $||x_{target} - x_{interrogator}||/c$. The power delay profile is generated according to the model described in the earlier sections; the peak power is calculated as indicated in (1) with path loss exponent $\alpha = 2.4$, while the exponential power delay profile is simulated with $\tau_{rms} = 10$ nS. The SNR at the reference distance as in (1) is $d_{ref} = 2$ space

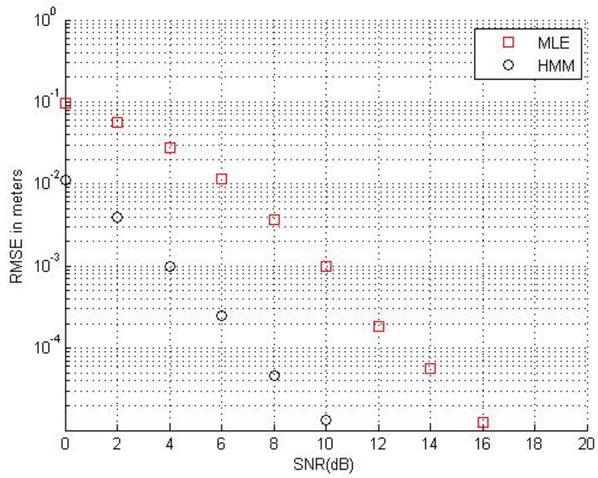

Fig. 4. Localization error

sample (i.e, 1m.) is $\eta_{ref} = 40$ dB. The algorithm performances are evaluated in terms of root mean squared error (RMSE) of location estimate $\tilde{q}_i$ as a function of spatial position x over a trajectory with 10000 steps that covers the whole area χ. For a given position $x \in \chi$, the estimate is computed as: $RMSE(x) = [|S(x)|^{-1} \sum_{i \in S(x)} ||q_i - \tilde{q}_i||^2]^{0.5}$, where $S(x) = [i : q_i = x]$ is the set of all time instants in which the target moves across χ and $|S(x)|$ is the number of times the location x was traversed. As indicated in Fig.7 the position RMSE versus the number of iterations is plotted. For the same localization scenario adopted a position steps of 1000 samples is generated at the same value of η_{ref}. These values compare the localization accuracy for MLE and the proposed HMM model. The error floor at a very low and large SNR is determined by finite value of time domain support of each measurement and spatial sampling interval, respectively. Simulation results suggest that the performance gain provided by the HMM model is around 7 dB in presence of path loss.

VI. Conclusion

In this paper we demonstrated the application of the HMM model to the problem of localization in cognitive UWB radar networks. Simulation results indicate an overall improvement in terms of spatial resolution and effective localization of a moving target. Such an algorithm can find its application in the centralized detection and tracking problem for cognitive radar networks. Since the RSS-TOA profile from all the interrogators can be collected at a central data fusion centre which can learn through the statistical data and develop an HMM. This would facilitate an intelligent selection of set of interrogators to be powered in subsequent time frames. This scheme is therefore applicable for UWB radar networks where the total transmitted power is a constraint in the system design.

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